TF 18-19

5. (a) "Maximizing the projection variance on a principal axis is equivalent to minimizing the errors due to projection"-Justify the clause in the context of Principal Component Analysis (PCA). **(10)**

(b) What is eigenvector and what is eigenvalue? How are they related to the PCA? How does covariance matrix help to determine the principal components? Briefly explain. **(20)**

TF 16-17 (Most likely excluding (a))

4 Let, a vector $\mathbf{x}$ with dimension D can be generated from any one of the k independent Gaussian distributions with identity co-variance matrix. The probability of selection of Gaussian distribution i is w_i where $\sum_{i=1}^{k} w_i = 1$ and the probability of generation of sample $\mathbf{x}_j$ is given as $P(\mathbf{x}_j \mid \boldsymbol{\mu}_i, \mathbf{I}) = \frac{1}{\sqrt{(2\pi)^D}} e^{\left(-\frac{1}{2}(\mathbf{x}_j - \boldsymbol{\mu}_i)^T (\mathbf{x}_j - \boldsymbol{\mu}_i)\right)}$

For this particular model answer the following questions

(a) Write a generative model for above scenario. **(5)**

Contd P/3

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(b) If $\mathbf{x}_j$ denotes a position vector within 3-dimensional co-ordinate system and there are 4 mixture components, calculate the total number of trainable parameter. **(10)**

(c) Write the four steps of the expectation-maximization algorithm to estimate trainable parameters for above model. **(20)**

TF 22-23

(b) For the univariate data below, based on current parameter settings, the $p_{ij} = P(C= i|x_j)$ values have already been calculated in the **E-step** for **mixture of 2 Gaussians**. Produce the model parameter values in the **M-step.** **(15)**

Data Id	Feature	$P(C=1\|x_j)$
x1	49	0.25
x2	53	0.50
x3	45	0.25
x4	45	0.20
x5	51	0.15

(b) Suppose you have N numbers $x_1, x_2, \ldots x_N$ drawn form a Gaussian distribution. Recall that the Gaussian density function on a single variable is: **(10)**

$$P(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

This linear-Gaussian model has two parameters: mean (μ) and standard deviation (σ). Derive the maximum-likelihood estimates of these two parameters.

TF 20-21

(b) Write down the pseudo-code of the k-means clustering method. How can you find out a suitable k for the above clustering method? **(8+6)**

TF 19-20

(b) What is a Gaussian mixture model? Write down and explain the equation for the probability of a data point under this model. How can you learn the parameters of this model using the EM algorithm? **(15)**

(c) What is a MAP hypothesis? What are its advantages and disadvantages over Bayesian learning? **(5)**